

NMI-style prediction/error matrix, top-50 as predicted positive

Cell class compares 2022 prediction with realized TG links in 2023-2026; number = hybrid score, parentheses = bridge count

iTE/TD donor concept

moisture-gradient ion transport	TP 0.94 (6)	TP 0.82 (3)	TP 0.75 (4)	TP 0.67 (3)			FP 0.90 (5)	TP 0.92 (5)			FP 0.83 (4)		TP 0.87 (7)				
cation/ anion thermodiffusion asymmetry	TP 0.88 (4)	TP 0.76 (3)	FP 0.69 (4)		TP 0.78 (4)	TP 0.79 (3)	FP 0.78 (3)	TP 0.72 (3)			FN 0.62 (2)		FN 0.66 (3)				
phase-transition entropy amplification		TP 1.00 (8)	TP 0.98 (8)		TP 0.99 (8)				TP 0.89 (6)	TP 0.87 (5)		FP 0.88 (5)					
anti-freezing organohydrogel design		TP 0.96 (7)			FP 0.79 (3)	TP 0.95 (6)		TP 0.97 (8)		TP 0.72 (3)	FP 0.72 (3)						
ion-pair complexation control				FP 0.92 (7)	TP 0.88 (6)		TP 0.96 (8)			FP 0.79 (5)	TP 0.79 (5)		FP 0.79 (5)		FN 0.33 (2)		
dynamic crosslinked gel network	TP 0.99 (10)	TP 0.88 (5)			FP 0.74 (4)			FP 0.93 (8)		FN 0.60 (3)	FN 0.63 (3)	TP 0.85 (6)			FN 0.10 (0)		
cellulose nanofiber scaffold	TP 0.97 (7)	TP 0.89 (4)	FP 0.88 (6)	FP 0.74 (4)								FP 0.79 (3)					
thermoresponsive micellization p/ n conversion	FP 0.82 (4)				FP 0.67 (4)				FP 0.70 (4)								
soft mixed ionic-electronic coupling	FP 0.75 (3)						FP 0.69 (3)										
high-entropy redox/ion coupling	TP 0.70 (3)	FN 0.50 (2)															
nanochannel-confined ion transport							FP 0.67 (3)		FN 0.53 (2)								
convection-resolved resistance decomposition											FN 0.60 (3)						
ion-dipole interaction transport								FN 0.51 (2)									

- TP: predicted and realized
- FP: predicted only
- FN: missed but realized
- TN: neither
- not a candidate pair

TG anchor concept